

1. Stereo disparity to 3D point

$$d(u,v) = u_L - u_R, \quad Z = \frac{f_x B}{d}$$
$$X = \frac{(u - c_x)Z}{f_x}, \quad Y = \frac{(v - c_y)Z}{f_y}, \quad \mathbf{p}_{\text{rect}} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

Rectified stereo converts image pixels into a semi-dense depth cloud.

2. Gravity-aligned frame

$$\mathbf{p}_g = R_{GI} R_{IC} R_{CR} \mathbf{p}_{\text{rect}}, \quad \mathbf{g} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

rectified camera → cam0 → IMU → gravity-aligned frame

IMU/EKF orientation supplies the vertical reference for plane classification.

3. RANSAC plane fitting

$$\pi : \mathbf{n}^\top \mathbf{p}_g + b = 0, \quad \|\mathbf{n}\| = 1$$
$$(\mathbf{n}^*, b^*) = \arg \max_{\mathbf{n}, b} \sum_i \mathbf{1}(|\mathbf{n}^\top \mathbf{p}_i + b| < \tau)$$

The best plane is the one with the largest inlier support.

4. Floor / wall decision

$$\text{floor} : \quad |\mathbf{n}_f^\top \mathbf{g}| > \cos \theta_f, \quad |\mathbf{n}_f^\top \mathbf{p}_i + b_f| < \tau_f$$
$$\text{wall} : \quad |\mathbf{n}_w^\top \mathbf{g}| < \sin \theta_w, \quad |\mathbf{n}_w^\top \mathbf{p}_i + b_w| < \tau_w$$
$$\text{label}(\mathbf{p}_i) = \begin{cases} \text{floor}, & \mathbf{p}_i \in I_f \\ \text{wall}, & \mathbf{p}_i \in I_w \\ \text{other}, & \text{otherwise} \end{cases}$$

Pipeline summary

$$\text{rectified stereo pair} \xrightarrow{\text{SGBM}} d(u,v) \xrightarrow{Z=f_x B/d} \mathbf{p}_{\text{rect}} \xrightarrow{R_{GI} R_{IC} R_{CR}} \mathbf{p}_g \xrightarrow{\text{RANSAC}} \{\text{floor}, \text{wall}, \text{other}\}$$

Key idea: floor normal $\parallel \mathbf{g}$, while wall normal $\perp \mathbf{g}$. Stereo provides depth; IMU provides the gravity axis.